
Environmental
Health
#IAmEnvironmentalHealth

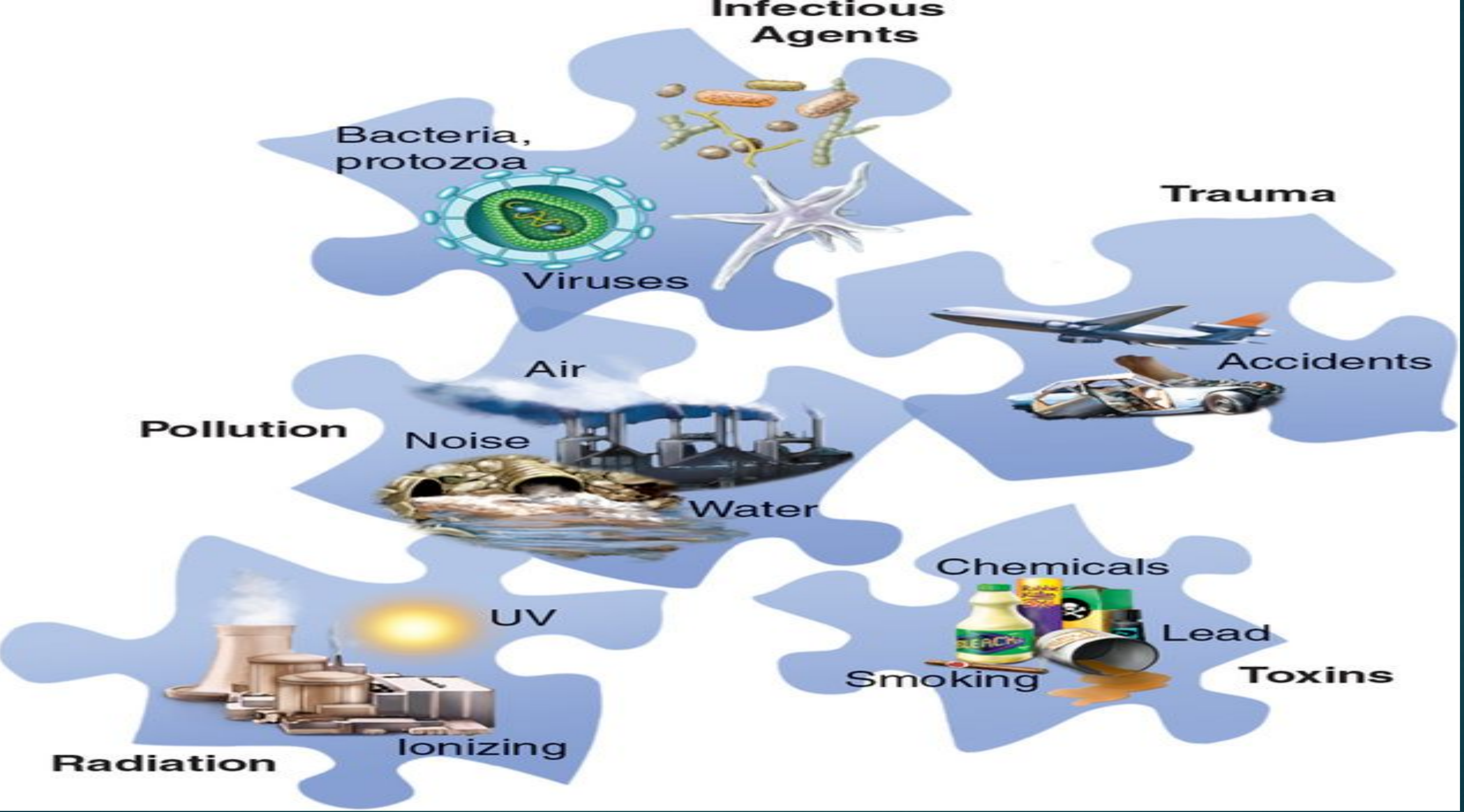
&

Part 1


TOXICOLOGY

Introduction

- Environmental Health and Toxicology is responsible for two distinct areas of public health concern as follows:-
- Identification of potential health hazards resulting from exposure to certain chemical or biological agents,
- Assessment and subsequent recommendations to abate or reduce any resulting health effects.


Infectious
Agents
Bacteria,
protozoa
Viruses
Trauma
Accidents
Air
Pollution
Noise
Water
Chemicals
UV
Lead
Smoking
Toxins
Ionizing
Radiation

Environmental toxicology

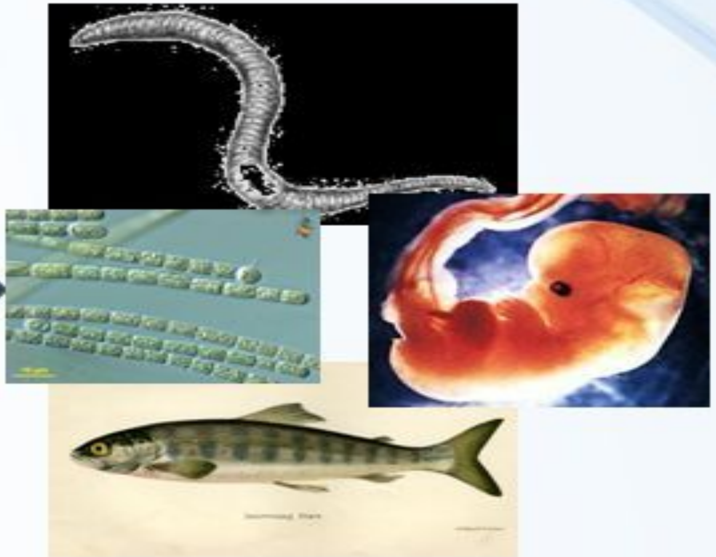
It is a multidisciplinary field of science concerned with the study of harmful effects of various chemical, biological and physical agents on living organisms. Humans, animals, and plants are increasingly being exposed to chemicals in the environment. The ever-increasing use of chemicals in industries has also resulted in further pollution of the environment. As toxic chemicals are widespread in the environment, there is a potential for these chemicals to cause significant damage and harmful effects on human health.

- Almost any chemical can be harmful depending on the **dose** (the amount of chemical and amount of time a person is exposed, in addition to how often a person is exposed) and **route of exposure** (whether the chemical is eaten, breathed in, or touches your skin). **Acute** (short-term) exposure and **chronic** (long-term) exposures can have different effects on a person's health.

CHEMICAL ENTERS THE ENVIRONMENT

LEVELS, FATE, PROCESSES

Bioavailable fraction



"EXPOSURE"

acute

chronic

CHEMICAL ENTERS THE ORGANISM

biomonitoring

<u>Toxikokinetics</u>

biotransformation
bioactivation
excretion / sequestration

Target site

"EFFECT"

Environmental hazard

Hazard is a situation that poses a level of threat to life, health, property, or the environment. An environmental hazard is an event that has the potential to threaten the surrounding natural environment or adversely affect people's health including pollution and natural disaster

- **Types of environmental Hazards:** hazards are classified into five
- **Physical**: occur naturally in our environment e.g. earthquakes, floods, volcanoes, drought, and fire outbreaks.
- **Cause**: due to deforestation, channelizing rivers (flooding), and deforesting slopes (landslides)
- **Reduce:** by better environmental choices, planting trees, avoiding excess cultivation of land

- **Chemical:** synthetic chemicals such as pesticides, disinfectants, and pharmaceuticals. Harmful natural chemicals also exist, including Urushiol(an oil in poison ivy, etc)
- **Biological:** biological substances from ecological interactions. Viruses, bacteria, and other pathogens.
- Infectious (communicable and transmissible) diseases, parasites, we can reduce the likelihood of infection
- **Cultural**: result from the place we live, our socio-economic status, our occupation, our behavioral choices
- **Psychosocial:** due to stress, hostility, depression, and hopelessness. This affects mental well being of a person.

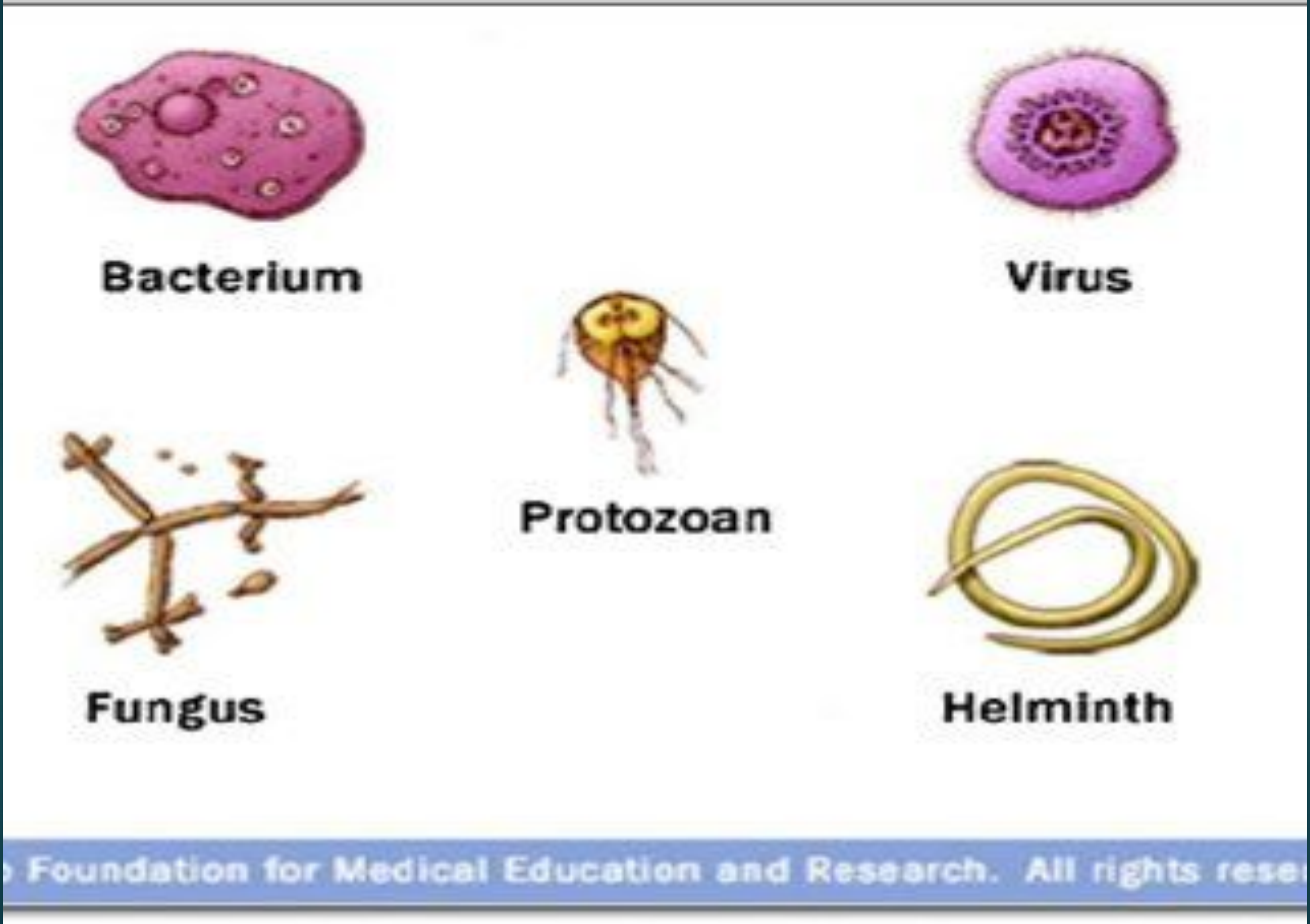
a)Physical hazard

(b) Chemical hazard

(c) Biological hazard

Infectious organisms

- otherwise called pathogens or disease-causing organisms, they are classified into five. Bacteria, viruses, Fungi, Protozoa, helminthes, and worms
- **Morbidity and quality of life:** any physical or psychological state considered to be outside the realm of normal well-being. The term **morbidity** is often used to describe illness, impairment, or degradation of health, especially when discussing chronic and age-related diseases. Morbidity doesn't necessarily mean that your ill health is immediately life-threatening. Over time, however, if an illness continues it may increase your risk of mortality (death). The most common are Heart disease, cancer, chronic lower respiratory diseases, stroke, Alzheimer's disease, diabetes mellitus, pneumonia and influenza, kidney disease, and suicide accounting for almost 75 percent of deaths in the U.S in 2013. In addition to infectious diseases, foodborne illness, associated infections and sexually transmitted diseases also contribute to higher morbidity among Americans.


Bacterium
Virus
Protozoan
Fungus
Helminth
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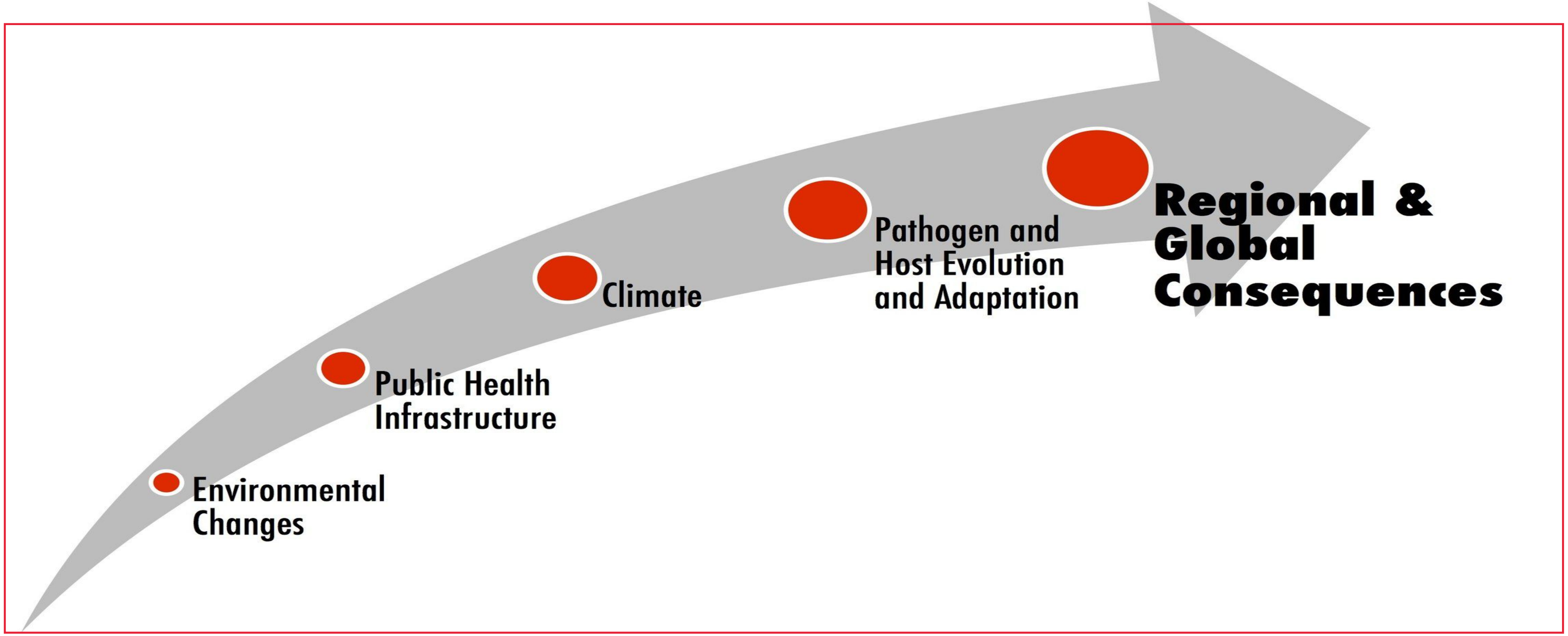
Safety and Prevention

Ways to lower morbidity rates include increasing screenings and early diagnosis which would lessen the length and impact of the disease on a person's quality of life. This would also lower complications and lower the mortality rates of certain diseases because early treatment is often the most effective. Other ways to compress morbidity are through education and access to preventive healthcare. For example, a model to lower morbidity for pregnant women involves access to safe abortions, prenatal care throughout pregnancy, and postpartum care along with family planning education throughout. As people live longer, the focus on lowering morbidity is education throughout life to create healthy habits and monitor health outcomes. A critical step is to receive regular check-ups to promote lifelong health before any signs of disease occur.


Emerging and Re-emerging
Diseases

Emerging disease due to environmental changes

- Environmental changes have a huge impact on the emergence of certain infectious diseases, mostly in countries with high biodiversity and serious unresolved environmental, social, and economic issues. An extensive review revealed a relationship between infectious disease outbreaks and environmental and climate change (heat wave, drought, flood, increased temp, high rainfall, etc) or environmental changes such as habitat fragmentation, deforestation, urbanization, and bush meat consumption. According to the World Health Organization, environmental threats to human health at global and regional levels include: "climate change, stratospheric ozone depletion, changes in ecosystems due to loss of biodiversity, changes in hydrological systems and supplies of freshwater, land degradation, urbanization, and stresses on food-producing systems" (WHO 2017). Rapid social and environmental changes are occurring globally, affecting both low-income countries and the largest advanced economies. Travel and transportation are the greatest increase in the risks of the rapid spread of infectious diseases (Jaffry et al. 2009). Particularly in the tropics where there is high biodiversity, ecological change is greatest, making these regions potential hotspots for the emergence of new pathogens affecting human, wildlife, and domestic animal health (Jones et al. 2008; Wilcox and Gubler 2005).

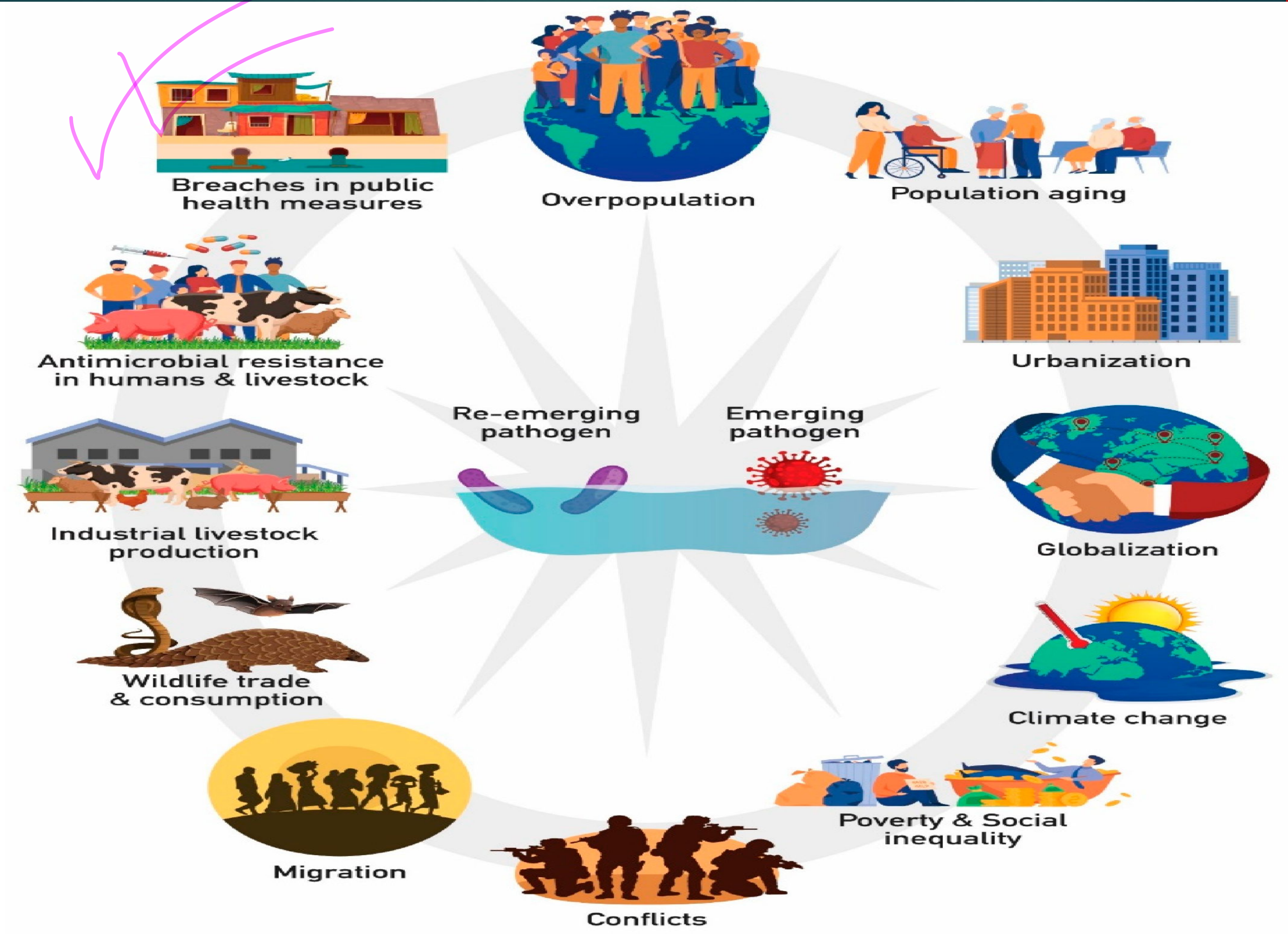


Wilcox, Bruce A., and Rita R. Colwell. "Emerging and reemerging infectious diseases: biocomplexity as an interdisciplinary paradigm." EcoHealth 2.4 (2005): 244.

Ecological diseases

- Emerging ecological diseases present a major threat to world health. On average, two new species of the human virus are reported each year, most having an animal origin. Recent examples are **SARS**, **swine flu**, and **avian influenza**. In 2014, the **Ebola virus** re-emerged from a bat reservoir, causing a major epidemic. Climate change could lead to ***Aedes*** **mosquitoes** establishing in New Zealand, and with them **dengue** and **Zika viruses**. The problem of disease emergence also extends to animal and plant populations. In most current epidemics, ecological factors (e.g. migration, climate, agricultural practices) play a more significant role in disease emergence than evolutionary changes in pathogens or hosts. Specifically, factors affecting the environment include the depletion of forests, expansion, and modernization of agricultural practices, and natural disasters such as floods. deliberate human action.

- These potentially lead to changes in microbial ecological diseases affecting human health. Sociodemographic factors such as an increase in population density, falling living standards, the decline of infrastructure, human travel, conflicts, social instability, and the killing of wild animals for meat all lead to an increase in emerging ecological diseases, which facilitate infections in humans. There are also some pathogens (disease causing organisms) whose emergence is as a result of deliberate human action.
- A reasonable public health response towards addressing emerging ecological disease problem in general aim at addressing **the fundamental factors that promote the occurrence and persistence of these diseases**, while embarking on **appropriate control measures**. WHO (world health organization) therefore supports **advocacy, awareness and pathogenesis studies, development and deployment of diagnostic tools, therapeutic drugs and vaccines** as the pillars of public health response


Breaches in public health measures
Overpopulation
Population aging
Antimicrobial resistance in humans & livestock
Urbanization
Re-emerging pathogen
Emerging pathogen
Industrial livestock production
Globalization
Wildlife trade & consumption
Climate change
Poverty & Social inequality
Migration
Conflicts

Toxic chemicals

The U.S. Environmental Protection Agency (EPA) defines a toxic chemical as any substance which may be harmful to the environment or hazardous to your health if inhaled, ingested or absorbed through the skin.

- **Toxic Chemicals in Your Home**
- Many useful household projects contain toxic chemicals. Common examples include:
- Drain cleaner
- Laundry detergent
- Furniture polish
- Gasoline
- Pesticides
- Ammonia
- Toilet bowl cleaner
- Motor oil
- Rubbing alcohol
- Bleach
- Battery acid. While these chemicals may be useful and even necessary, it is important to remember they should be used and disposed of according to instructions on the packaging.

Types of Toxins

- Toxins may be categorized into four groups. It's possible for a substance to belong to more than one group.
- **Chemical Toxicants** - Chemical toxins include both **inorganic substances**, such as mercury and carbon monoxide, and organic compounds, such as methyl alcohol.
- **Biological Toxins** - Many organisms secrete toxic compounds. Some sources consider pathogenic organisms to be toxins. A good example of a biological toxin is tetanus.
- **Physical Toxicants** - These are substances that interfere with biological processes. Examples include asbestos and silica.
- **Radiation** - Radiation has a toxic effect on many organisms. Examples include gamma radiation and microwaves


TIN POISONING
FLUORIDE
DIESEL
BPA
(BISPHENOL A)
LEAD
FOOD PESTICIDES
AND CHEMICALS
TOXINS
MERCURY
OVER 4000 CHEMICALS
ALUMINUM,
PARABENS
DRY
CLEAN
PERC
(PERCHLOROETHYLENE)
FLAME RETARDANTS
INSECT PESTICIDE
AMMONIA,
MANGANESE
EXHAUST AND
POLLUTION

Natural Toxic Chemicals

Many toxic chemicals occur in nature. For example, plants produce toxic chemicals to protect themselves from pests. Animals produce toxins for protection and to capture prey. In other cases, toxic chemicals are simply a by-product of metabolism. Some natural elements and minerals are poisonous. Here are some examples **of natural** toxic chemicals

- Mercury
- Snake venom
- Caffeine in coffee, tea, kola and cocoa
- Arsenic
- Ricin from castor beans
- Petroleum
- Hydrogen sulfide
- Chlorine gas
- Smoke

Industrial and Occupational Toxic Chemicals

The US Occupational Safety and Health Administration (OSHA) have identified several chemicals it considers highly hazardous and toxic. Some of these are laboratory reagents, while others are used commonly in certain industries and trades. Certain pure elements are included. Here are a few substances on the list (which is extremely long):

- Acetaldehyde
- Acetone
- Acrolein
- Bromine
- Chlorine
- Cyanogen
- Isopropyl alcohol
- l-limonene
- Hydrogen peroxide >35%

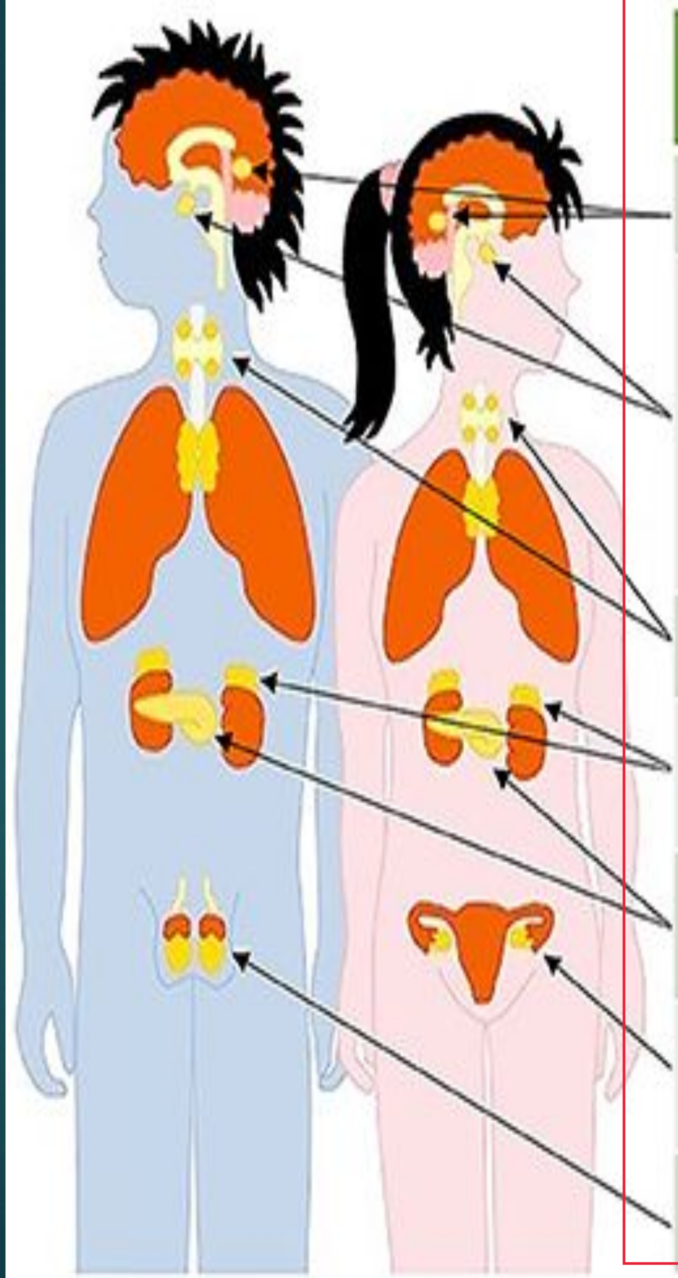
Are All Chemicals Toxic?

Labeling a chemical as "toxic" or "non-toxic" is misleading because any compound can be toxic, depending on the route of exposure and the dose. For example, **even water is toxic** if you drink enough of it. Toxicity depends on other factors besides dose and exposure, including species, age, and gender. For example, humans can eat chocolate, yet it's toxic to dogs. In a way, all chemicals are toxic. Similarly, there is a minimum dose for nearly all substances below which toxic effects are not seen, called the toxicity endpoint. A chemical can be both necessary for life and toxic. An example is iron. Humans need low doses of iron to make blood cells and perform other biochemical tasks, yet an overdose of iron is deadly. Oxygen is another example.

Hormones: The chemical messengers

The endocrine system is made up of many glands throughout your body, including the pituitary gland, ovaries or testes, and thyroid. These glands secrete hormones that enter your circulatory system and bind with receptors in other parts of the body, signaling to your organs and tissues what to do and when to do it. Hormones help regulate many bodily functions, including:

- Growth and development
- Blood sugar control
- Metabolism and energy
- Blood pressure
- Reproductive processes
- Appetite and weight control.
- Digestion
- Sleep cycles
- Mood

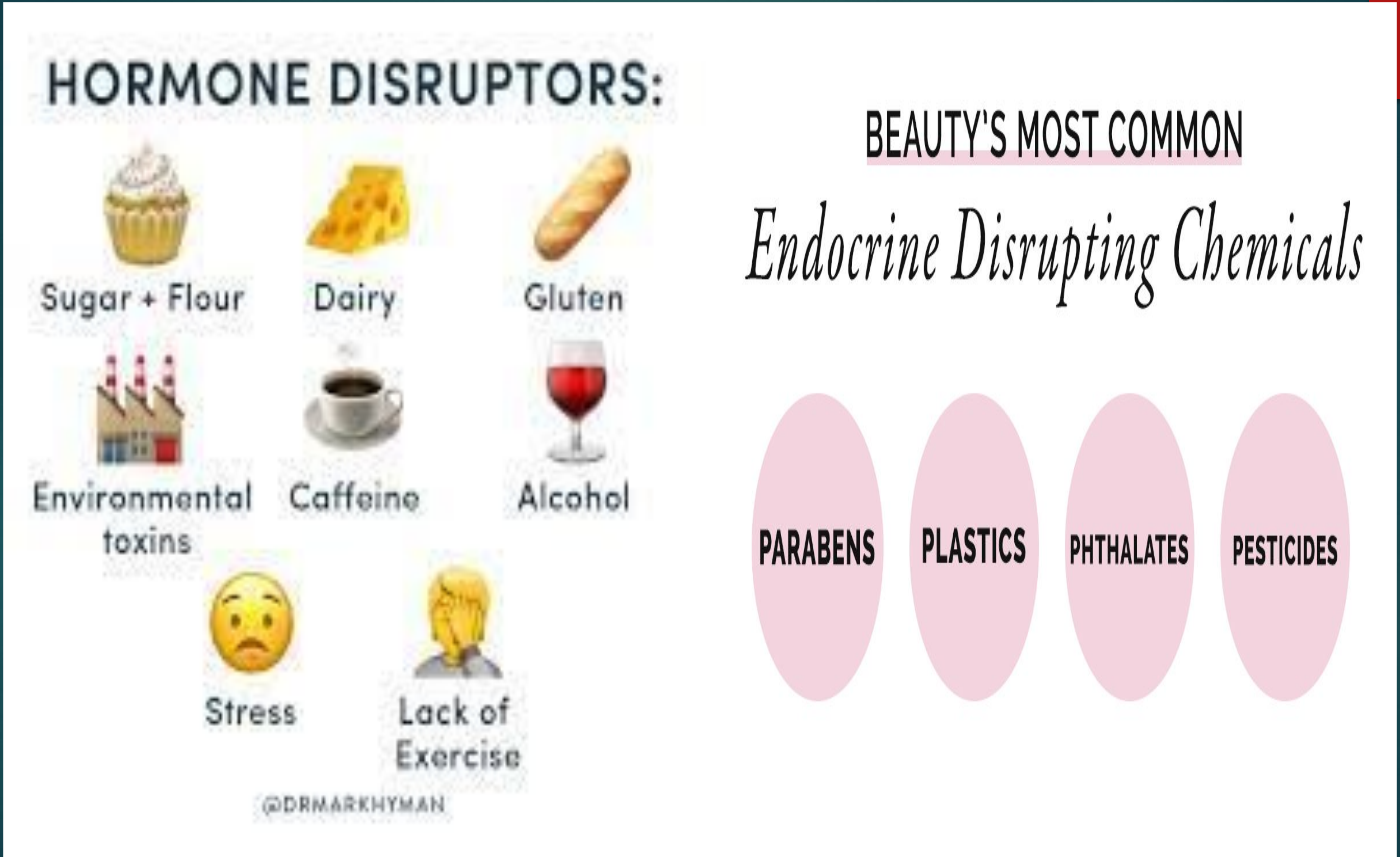


Gland	Hormone	Target Organ	Function
Pineal gland	Melatonin	Many	Biological clock
Pituitary gland	FSH / LH ADH Growth hormone Oxytocin Prolactin	Ovaries Kidneys Uterus Breast tissue Many others	Menstrual cycle Osmoregulation Growth & division Birth contractions Milk production
Thyroid gland	Thyroxin	Liver	Metabolic rate
Adrenal glands	Adrenaline Cortisol	Many	Fight or flight Anti-stress
Pancreas	Insulin Glucagon	Liver	Blood sugar levels
Ovaries	Estrogen Progesterone	Uterus	Menstrual cycle
Testes	Testosterone	Many	Male characteristics

Hormone disrupters

These are manmade chemicals that can interfere with your body's hormone functioning. These are hundreds of chemicals thought to have hormone-disrupting effects. Every day we come in contact with chemicals (including parabens and phthalates) that expert think, over time, could mess with how our body's hormones function. These are called hormone- or endocrine disrupting chemicals (EDCs). They're in food packaging, household and personal care products, and even in the air that we breathe "they are essentially everywhere, we are consuming them, we are putting them on our bodies, we are being exposed to them, and most of the time we don't even know it". Below are some categories of hormone disrupters:-

- **Bisphenols, including bisphenol A (BPA):** BPA is found in food beverage packaging, adhesives and many other consumer products. It can leach into foods and beverages that we consume or enter the body through the skin. Studies show that most people have at least some BPA in their body.
- **Phthalates:** are used to make plastics, or as dissolving agents for other materials. They are found in medicinal tubing, detergents, automotive parts, cosmetics, and other products that we touched every day.
- **Polychlorinated biphenyls (PCBs):** these chemicals have been banned in the USA since late 1970s but were widely used for decades prior to that in electrical equipment's, paints, dyes, plastics and rubber products. They remain in the environment.
- **Dichlorodiphenyltrichloroethane (DDTs) and dichlorodiphenyldichloroethylenes (DDEs):** these are also environmental pollutants. DDE is a breakdown product of DDT which is banned in the U.S but still used in other countries as pesticide.


HORMONE DISRUPTORS:
Sugar + Flour
Dairy
Gluten
Environmental toxins
Caffeine
Alcohol
Stress
Lack of Exercise
@DRMARKHYMAN
BEAUTY'S MOST COMMON
Endocrine Disrupting Chemicals
PARABENS
PLASTICS
PHTHALATES
PESTICIDES

Here are some tips for minimizing your exposure

- Consider choosing products with BPA- and phthalate-free packaging.
- Drink tap water that has been filtered.
- Avoid microwaving plastics.
- Avoid personal care products that list “fragrance” as an ingredient, as phthalates often hides in fragrances.
- Choose products labeled “phthalate-free,” “paraben-free” or “BPA-free.”
- Avoid unnecessary exposure to industrial chemicals and pesticides.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Air
Photolysis
Oxidation
Precipitation
Source
Industry
Agriculture
Domestic, etc.
Biota
Metabolism
Storage
Excretion
Soil and Sediment
Photolysis and
metabolism
Evaporation
Water
Hydrolysis
Oxidation
Microbial degradation
Evaporation
Sedimentation


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Water
(drinking)
Cosmetics,
household accidents
(incidental and
accidental exposu
Medicinal drugs
(oral, intravenous,
intramuscular)
Food
(eating)
Air
(breathing)
Occupational exposure
(oral, dermal, inhalation)

Diet

- In nutrition, **diet** is the sum of food consumed by a person or other organism. The word diet often implies the use of specific intake of nutrition for health or weight-management reasons (with the two often being related). Although humans are omnivores (ability to eat and survive on both plant and animal matter, obtaining energy and nutrients from plant and animal, each culture and each person holds some food preferences or some food taboos. This may be due to personal tastes or ethical reasons. Individual dietary choices may be more or less healthy.
- Complete nutrition requires ingestion and absorption of vitamins, minerals, essential amino acids from protein, and essential fatty acids from fat-containing food, also food energy in the form of carbohydrate, protein, and fat. Dietary habits and choices play a significant role in the quality of life, health and longevity (life expectancy).

A healthy diet is essential for good health

- It protects you against many chronic non communicable diseases, such as heart disease, diabetes and cancer. Eating a variety of foods and consuming less salt, sugars, saturated and industrially-produced trans-fats, are essential for healthy diet. A healthy diet comprises a combination of different foods. These include:
- Staples like cereals (wheat, barley, rye, maize or rice) or starchy tubers and roots (potato, yam, taro or cassava).
- Legumes (lentils and beans).
- Fruit and vegetables.
- Foods from animal sources (meat, fish, eggs and milk).

Content….

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Topic 2: The Global Environmental Picture

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Topic 4: Air Pollution

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Topic 7: Municipal Waste Management

Topic 8: Energy Resource

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Topic 10: Environmental legislation of Bangladesh

Topic 11: Emerging Environmental Issues

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Will cover….**Final exam**